



Mathematical Modeling of Influenza Dynamics: Integrating Seasonality and Gradual Waning Immunity

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Abstract

The dynamics of influenza virus spread is one of the most complex to model due to two crucial factors involved: seasonality and immunity. These factors have been typically addressed separately in mathematical modeling in epidemiology. In this paper, we present a mathematical modeling approach to consider simultaneously both forced-seasonality and gradual waning immunity. A seasonal SIR_n model that integrates seasonality and gradual waning immunity is constructed. Seasonality has been modeled classically, by defining the transmission rate as a periodic function, with higher values in winter seasons. The progressive decline of immunity after infection has been introduced into the model structure by considering multiple recovered subpopulations or recovery states with transmission rates attenuated by a susceptibility factor that varies with the age of infection. To show the applicability of the proposed mathematical modeling approach to a real-world scenario, we have carried out a calibration of the model with the data series of influenza infections reported in the 2010–2020 period at the General Hospital of Castellón de la Plana, Spain. The results of the case study show the feasibility of the mathematical approach. We provide a discussion of the main features and insights of the proposed mathematical modeling approach presented in this study.

Keywords Mathematical modeling · Influenza · Seasonality · Gradual waning immunity · Susceptibility

Mathematics Subject Classification 92D30 · 92-10

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1 Introduction

Influenza is one of the most prevalent and recurrent diseases worldwide. Due to its seasonal nature, around one billion cases are affected annually, including 3–5 million cases of severe illness, and it causes from 290 000 to 650 000 deaths due to its severe damage to the respiratory system [1, (Lofgren et al. 2007; Neumann and Kawaoka 2022; Tang and Loh 2016)]. It is known that this disease is caused by different strains of the influenza virus. However, from an epidemiological point of view, the transmission dynamics of the influenza virus occur by multiple factors, which makes its modeling and study very challenging (World Health Organization 2024; Basile et al. 2018; Brugger and Althaus 2020; Endo et al. 2021; Hill et al. 2019; Moorthy et al. 2012; Sara et al. 2020). Thus, despite many studies devoted to investigating the influenza dynamics in humans the understanding of it remains incomplete (Chen et al. 2022; Dalziel et al. 2018; Kenah et al. 2011; Magal et al. 2020; Moorthy et al. 2012; Tamerius et al. 2011; Zhang 2015).

One of the factors that explain influenza seasonal behavior every year is the high mutation capacity of the influenza virus (Liu and Walker 2023; Neher et al. 2016; Shi et al. 2010). Due to its viral RNA structure, which is more unstable and easily mutable than DNA, and its ability to colonize zoonotic reservoirs, which allows influenza to persist throughout the year in the population, influenza generates new variants every year (Webster and Govorkova 2014; Barr and Fearn 2016). This means that they are able to evade naturally acquired immunity or vaccines (i.e., immuno-escape) and maintain their infectious capacity in the human population. We also know that influenza is especially prevalent in winter seasons, where cold and humidity are factors that facilitate the infection (Lowen and Steel 2014). For instance, recently it has been found that the antiviral immune responses are weakened by cold exposure (Huang et al. 2023). The explanation of this is that cold exposure decreases the total extracellular vesicle secretion as well as diminishes microRNA packaging and antiviral binding affinity of individual extracellular vesicles. In addition, some viruses replicate better at lower temperatures than at the natural body temperature (Papadopoulos et al. 1999). There are other works that have found underlying molecular mechanisms that show that cold temperature may diminish the host's innate immune response to viral infections (Foxman et al. 2015, 2016).

Another highly influential factor in seasonal influenza epidemic waves and their magnitude is the immunity against the virus. When a person is exposed to a virus and develops an immune response, this person acquires immunological memory (Combadière et al. 2010; Cox et al. 2004; Kim et al. 2018; Olmos Liceaga et al. 2023). It is well-known that immunity varies depending on many factors including the host immune response of each individual and the time since the last infection, which is related to the waning effect of immunity (Krammer 2019). It is known that the host immune response affects the severity and duration of influenza infection or other type of virus infections (Barrett et al. 2021; Dobrovolny et al. 2013). Moreover, the susceptibility can vary depending on the virus strain to which the host is exposed, but also from past exposures to the same or similar strains, which may confer the so-called cross-immunity. (Ahmed et al. 2007; Barrett et al. 2021; Cobey and Hensley 2017).

A large part of the population acquires immunity through having been infected, while another part of the population acquires it thanks to the vaccine, as a preventive action to be protected against the virus. It is generally known that natural immunity lasts longer than vaccine immunity and protects against more types of strains (thanks to the cross-immunity) (Epstein and Price 2010; Krammer 2019). Regarding the effects of vaccination immunity on influenza transmission, only some part of the population is directly protected by vaccination immunity due to vaccination coverage. This coverage depends on many variables such as the vaccine's economic cost and availability, vaccination campaign strategies, the population's willingness to be vaccinated, etc. (Tracht et al. 2010; Arda et al. 2011; Castillo-Rodríguez et al. 2022; Imai et al. 2018; Putri et al. 2018). Additionally, it should be noted that the vaccine's ability to confer immunity (its efficacy) is usually partial, and not necessarily against all circulating virus strains. In contrast, the natural immune response has several components and mechanisms of action that are not yet very well understood (Krammer 2019; Patel et al. 2021). The immune response can be divided into two main pieces; the adaptive and the innate response. A key property of the adaptive response is its memory (Kubo et al. 2017; Ratajczak et al. 2018). This memory helps to fight viral infections since it mounts a quick and specific response that recognizes an antigen that the body has previously encountered (Natoli and Ostuni 2019; Ratajczak et al. 2018). It has been mentioned that the immune memory plays an important role in the severity of an influenza infection (Azambuja 2010; Doherty et al. 2006). Cross-immunity is strongly related to the memory immune response which mainly originates from previous exposure to different viruses, and it plays an important role in repeated infections, pathogen diversity and mutations (Abu-Raddad and Ferguson 2004). As a particular case of the cross-immunity, during the well-known Spanish influenza pandemic of 1918, there were more than 17 million deaths (Ahmed et al. 2007; Noymer and Garenne 2000). Interestingly, 30-60 years old adults presented a lower mortality rate in comparison with younger adults aged 18-30 years. It has been argued that this happened due to factors related to the immunological memory (Ahmed et al. 2007; Gagnon et al. 2013, 2015; McAuley et al. 2015).

Based on all the previous epidemiological aspects, the mathematical modeling of seasonal influenza transmission dynamics is challenging and complex, especially if both seasonality or immunity factor are to be taken into account. Different mathematical models have been presented to study the dynamics of influenza epidemics and pandemics considering these two key factors, although the most widespread has been the compartmental models based on differential equations (Brauer et al. 2012; Yuan et al. 2021).

On the one hand, some mathematical models for influenza include a seasonally-forced function – classically, a time-dependent sinusoidal infection rate with a 1-year period – in order to generate a seasonal behavior in the infected population (Gabrick et al. 2024; Towers and Feng 2009). In Truscott et al. (2012), a model with a seasonally-forced function was used to estimate that a seasonal peak basic reproduction number is in the range of 1.6-3. In addition, it was suggested that the basic reproduction number depends on the timescale for the waning immunity to the circulating influenza strain, which was estimated to be between 3 and 8 years. Also, it was found that seasonal variation in transmissibility is in the range of 15-30% of its mean value. Another recent

work that deals with seasonal influenza and the estimation of the basic reproduction number is presented in Jing et al. (2020). The authors used a seasonal forced function and took into account meteorological factors including unreported cases. Nevertheless, seasonal behavior can also occur without a seasonally-forced function (Hethcote et al. 1981).

On the other hand, in compartmental models, immunity decay has been modeled by modifying the classical model equations into integro-differential equations (Bhattacharya and Adler 2012). The well-known SIR and SEIR epidemic model structures have been used to model waning immunity with regard to the influenza virus (Ehrhardt et al. 2019; Takeuchi and Kuroda 2010). Previous works have studied influenza by means of a variety of mathematical models that include gradual waning immunity explicitly. All these works provide important insights into the dynamics of seasonal influenza. For instance, in Goeyvaerts et al. (2015), a SEIRS mathematical model with age stratified classes and vaccination was presented. Each year some proportion of the susceptible population is seeded into the population as newly infectious individuals and a seasonal forced function was used. Despite some problems with model identifiability due to the correlation of several parameters, it was highlighted the importance of influenza seasonal variation. In El Khalifi and Britton (2023); Hethcote et al. (1981); Guiaş (2023), the authors divide the susceptible population into multiple stages with different immunity levels in order to simulate the immunity decay after infection. In Hethcote et al. (1981), a mathematical model that considers multiple recovered classes, but with the possibility of reinfection only in the last recovered class is constructed and analyzed. In El Khalifi and Britton (2023), a mathematical model that considers several recovered stages but no forced-seasonality in the transmission rate has been presented.

Regarding mathematical models for influenza that have included forced seasonality and explicit gradual waning immunity together, there are few works. In Dafilis et al. (2014), two recovered classes are included in order to consider gradual waning immunity but in the last recovered class people can transit to the completely susceptible population. In Xiao and Moghadas (2013), a mathematical model that includes impulse vaccination (even distribution) before each season was presented. The main simulation result was that the pattern of large seasonal epidemics is strongly correlated with the duration of specific cross-protection immunity induced by natural infection. The possibility of moving from the last recovered class to susceptible is considered in the model. Their study included a large range of simulations to explore the effect of several parameters. In Lai and Gog (2024), a new model based on a discrete system was presented in order to show that the waning immunity can produce sustained oscillations without using any seasonally-forced function.

In this paper, we propose a mathematical modeling approach to describe the dynamics of seasonal influenza combining forced-seasonality and gradual waning immunity. The proposed compartmental model – the seasonal SIR_n model – considers that the population that have been infected will transit through multiple recovered stages. In each stage, the individuals have a different susceptibility (or immunity) level, which modulates the infection rate of the individuals in the recovered stages. In particular, the susceptibility level (and thus, the probability of infection) increases as individuals progress through the recovered stages. In addition, the mathematical modeling

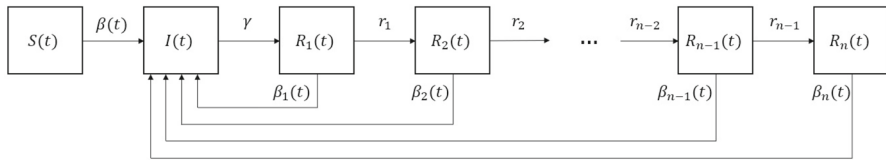


Fig. 1 Model flow chart of the seasonal SIR_n influenza epidemic model

approach takes into account the seasonality of the influenza virus by considering the infection rate as a 1-year periodic sinusoidal function. Some previous models (Hethcote et al. 1981; El Khalifi and Britton 2023) can be seen as good foundations for the model presented in this work. In order to show the applicability and reliability of the constructed mathematical modeling approach, we consider a case study by using the data series of influenza infections reported in the 2010–2020 period at the General Hospital of Castellón de la Plana city, Spain. We carry out a calibration process to fit the seasonal SIR_n model to the real data. In addition, a brief practical identifiability analysis of the unknown model's parameters is conducted for the case study.

The paper is structured as follows. In Section 2 we present in detail a complete description of the seasonal influenza mathematical model, including a biological interpretation of its equations and parameters. Section 3 is devoted to show the applicability and reliability of the proposed mathematical modeling approach by presenting a real-world case study. Finally, discussion and conclusions are presented in Sections 4 and 5, respectively.

2 Mathematical model

In this section, we present the foundation and main features of the proposed mathematical modeling approach that couples seasonality and immunity in influenza epidemiological dynamics. In particular, we present the seasonal influenza mathematical model, including a description of its equations and parameters. First, let us proceed to present the seasonal SIR_n model.

2.1 Mathematical model description

The constructed mathematical model, which we called the seasonal SIR_n model, assumes that the entire population $N(t)$ at time t is divided into $n + 2$ subpopulations: the susceptible $S(t)$, the infectious $I(t)$ and the n consecutive recovered $R_1(t), \dots, R_n(t)$, where $n \in \mathbb{N}$, $n \geq 2$. In contrast to the classical epidemiological models, where only one population of recovered individuals $R(t)$ appears, this model divides the recovered individuals into multiple stages according to their immunity level, which depends on the age of infection. The transitions between the multiple recovered stages enable the immunity progression of the individuals. The mathematical model schema is shown in Figure 1.

The natural process of infection due to influenza virus is the following:

1. A susceptible individual becomes infected due to a contact with other infected individuals. The decrease in susceptibles $S(t)$, or equivalently, the increase of infecteds $I(t)$ over time due to contact transmission is proportional to the number of possible contacts between susceptibles and infected $S(t)I(t)$.
2. The transmission rate $\beta = \beta(t)$ determines what fraction of all possible contacts are effective contacts, resulting in infection, and addresses all factors affecting contagion.
3. An infected individual becomes recovered after an infectious period of $\frac{1}{\gamma}$ days. It is estimated that the influenza infection time is between 5 and 7 days [69]. In this study, we assume $\gamma = 1/7 \text{ days}^{-1} = 1 \text{ weeks}^{-1}$.
4. A recovered person, immediately after infection, has some level of susceptibility to the virus. However, as time passes after infection, he/she progressively loses immunity. Approximating this phenomenon by a discrete process, this individual will transition through n recovered stages, each with a lower immunity level. The time in which the individual is in an R_k recovered stage before moving to the R_{k+1} stage is $\frac{1}{r_k}$ days. The value of r_k depends on how many and in what form the recovery stages are defined. Additionally, in each of these R_k recovered stages, there is a possibility that individuals will be re-infected by contact with infected individuals. Thus, the decrease of individuals in the recovered stage $R_k(t)$ is proportional to the number of contacts between recovered and infected $R_k(t)I(t)$, with a transmission rate $\beta_k(t)/N$, which depends on the contagion factors, and on the immunity level of the recovered individuals.
5. All populations are affected by total mortality (i.e. natural and infection-induced mortality), which occurs proportionally to the size of each population with a constant death rate μ . In contrast, births occur only in the susceptible population, since we assume that all those born do not have immunity to the influenza virus. We also assume that newborns are proportional to the entire population $N(t)$ with a constant birth rate λ (Hethcote 2000).

The mathematical model is given by the following non-autonomous system of nonlinear first-order ordinary differential equations:

$$\begin{cases} \dot{S}(t) &= \lambda N(t) - \mu S(t) - \beta(t) \frac{S(t)I(t)}{N(t)}, \\ \dot{I}(t) &= \beta(t) \frac{S(t)I(t)}{N(t)} + \sum_{k=1}^n \beta_k(t) \frac{R_k(t)I(t)}{N(t)} - \gamma I(t) - \mu I(t), \\ \dot{R}_1(t) &= \gamma I(t) - \beta_1(t) \frac{R_1(t)I(t)}{N(t)} - r_1 R_1(t) - \mu R_1(t), \\ &\vdots \\ \dot{R}_k(t) &= r_{k-1} R_{k-1}(t) - \beta_k(t) \frac{R_k(t)I(t)}{N(t)} - r_k R_k(t) - \mu R_k(t), \quad \forall k = 2, \dots, n-1 \\ &\vdots \\ \dot{R}_n(t) &= r_{n-1} R_{n-1}(t) - \beta_n(t) \frac{R_n(t)I(t)}{N(t)} - \mu R_n(t), \\ \dot{N}(t) &= \dot{S}(t) + \dot{I}(t) + \sum_{k=1}^n \dot{R}_k(t) \equiv \dot{N}(t) = (\lambda - \mu)N(t), \end{cases} \quad (1)$$

with $S(0), I(0), R_k(0), N(0) \geq 0$, $\forall k = 1, \dots, n$, as initial conditions, and where $\lambda \in \mathbb{R}_+$ is the birth rate, $\mu \in \mathbb{R}_+$ is the death rate, $\gamma \in \mathbb{R}_+$ is the recovery rate,

$\beta(t) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the transmission rate at time t for the S susceptible subpopulation, $\beta_k(t) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $k = 1, \dots, n$ is the transmission rate at time t for the R_k recovered subpopulation, and $r_k \in \mathbb{R}_+$, $k = 1, \dots, n - 1$ is the transition rate between R_k and R_{k+1} recovered subpopulations due to the waning immunity of the recovered individuals.

2.2 Immunity and susceptibility

As it has been previously explained, people who have been exposed to the influenza virus acquire a memory in their immune system, also known as natural immunity. However, this immunity is progressively lost or is partial against other strains of the virus. Thus, the individual becomes less immune, or in other words, more susceptible to becoming re-infected with the virus. In this study, in order to analyze the dynamics of seasonal influenza at the human population level, we consider that the infection of individuals can occur against all existing circulating variants, without distinguishing between them in the proposed model. Although the predominance of one strain over another may indeed change between seasons (due to the mutation of the virus and also to the progressively loss of immunity), this predominance alternates in a complex way between strains. Thus, our interest lies in modeling the immunity of individuals against all these strains on average and in the medium to the long term.

Additionally, an important aspect to consider is the effect of acquired immunity from vaccination. Individuals vaccinated in previous years are distributed in the recovered classes $R_k(t)$, as they have a certain level of immunity that varies according to the circulating influenza variants (Chan et al. 2018; Furuse and Oshitani 2016; Li et al. 2016). Explicit modelling of vaccination has been introduced in similar models by allowing transitions (connections) between recovered classes $R_k(t)$, allowing an individual to regain higher levels of immunity upon vaccination (El Khalifi and Britton 2023). In our case, as we are working with a more simplified model (since we do not have vaccination data), the effect of vaccine-induced immunity will be captured by the model parameters in the calibration process. In particular, vaccine immunity will result in a lower value of the beta infection rate, because the vaccinated population hinders the transmission and spread of the virus in the overall population.

The constructed SIR_n model captures the immunity effect introducing multiple recovered stages $R_k(t)$, $k = 1, \dots, n$, according to the individuals' immunity/susceptibility level. Here, we define the susceptibility level $s(\tau) : \mathbb{R}_+ \rightarrow [0, 1]$ as the relative ability of an individual to become infected at a time after infection τ with respect to a fully susceptible individual. It is important to note that the time after infection τ is different and independent from the time t of the model. The susceptibility function $s(\tau)$ is bounded between 0 (not susceptible/full immunized) and 1 (fully susceptible or, equivalently, not immunized). By this definition, immunity can be defined as $i(\tau) = 1 - s(\tau)$. Because of the natural immunity decay, it is reasonable to define $i(\tau)$ as a monotonic decreasing function and, consequently, $s(\tau)$ as a monotonic increasing function (Ranjeva et al. 2019).

Since immunity decay is a continuous process over time τ , there are infinite phases of susceptibility, and therefore, of recovered stages. It follows that, in order to design

a compartmental model such as the one proposed here, we have chosen a partition

$$\mathcal{T} = \{\tau_1, \dots, \tau_n\}$$

of different τ_i times in the $R_+ = [0, +\infty)$ interval, so that we obtain a set of susceptibility levels

$$\mathcal{S} = \{s(\tau_1), \dots, s(\tau_n)\}.$$

The partition can be arbitrarily chosen, although it is essential that it be sufficiently refined to faithfully represent the susceptibility increase process given by the $s(\tau)$ susceptibility function shape. In this way, the n recovered subpopulations or states $R_k(t)$ are defined, each with its level of susceptibility $s(\tau_k)$, $\forall k = 1, \dots, n$. The n parameter is similar to the number of stages used in the gamma distributed epidemiological models, and gives to the recovered phase a more realistic distribution than the classical exponential distribution that is oftentimes used in epidemiological models (Best and Perelson 2018; González-Parra et al. 2018; Lloyd 2001; O'Neill and Becker 2001). Alternatively to this approach, an integro-differential system can be constructed using a continuous susceptibility function $s(\tau)$ (Hethcote et al. 1981; Hethcote 2000). In Brauer et al. (2019) it has been mentioned that it is possible to assume that at the beginning of a seasonal influenza season individuals have some level of cross-immunity that depends on the previous seasonal influenza strains, up to a maximum of n seasons as we have assumed in the proposed seasonal SIR_n model.

2.3 Transition and transmission rates

Having defined the partition $\mathcal{T}$, the transition rate r_k between the $R_{k-1}(t)$ and $R_k(t)$ recovered subpopulations can be defined as

$$r_k = \frac{1}{\Delta\tau_k} = \frac{1}{\tau_{k+1} - \tau_k}, \quad \forall k = 1, \dots, n-1. \quad (2)$$

Note that these transition rates are equal if the partition $\mathcal{T}$ is uniformly divided.

Regarding the transmission rates, $\beta(t)$ and $\beta_k(t)$, these are related, as in classical models, with the effective contact rates between individuals (with the probability of an individual becoming infectious after an effective contact), but also with the susceptibility level of the individuals. Using the susceptibility function, the transmission rate $\beta_k(t)$ for the $R_k(t)$ recovered subpopulation can be defined as

$$\beta_k(t) = s(\tau_k)\beta(t) = [1 - i(\tau_k)]\beta(t), \quad k = 1, \dots, n. \quad (3)$$

With this definition, note that, if the susceptibility function $s(\tau)$ is monotonically decreasing, then one gets that $\beta_k < \beta_{k+1}$, $\forall k = 1, \dots, n-1$. As particular cases of interest,

- Typically, $\tau_1 = 0$ and $s(\tau_1) = s(0) \simeq 0$, so that $\beta_1 = s(0)\beta(t) = 0$, i.e., individuals after infection in R_1 population are completely immunized and cannot be infected (R_1 is the classical R recovered population).

- If $s(\tau_n) \simeq 1$, then $\beta_n = s(\tau_n)\beta(t) \simeq \beta(t)$, i.e., after a long period of time following infection, individuals in R_n population are almost completely susceptible to reinfection.

Thus, based on the previous aspects to fully characterize the influenza infection dynamics with the seasonal SIR model, it is sufficient to define γ , $\beta(t)$, $s(\tau)$ (or alternatively, $i(\tau)$), and the partition $\mathcal{T}$.

2.4 Seasonality

With regard to the seasonality, we know that there are infections that depend on the season of the year, since there are several factors at that season (temperature, humidity, contacts, human behavior, etc.) that contribute to the spread of the pathogens (Foxman et al. 2015, 2016; Huang et al. 2023; Lowen and Steel 2014; Papadopoulos et al. 1999). Usually, the winter seasons are those in which seasonal influenza waves occur (Ewing et al. 2017; Greiff et al. 2021; González-Parra et al. 2011; Hill et al. 2019; Ho et al. 2019). One of the classic ways of modeling seasonality in epidemiological mathematical models is to define the transmission rate as a time-dependent function $\beta(t)$ (Arenas et al. 2021; Grassly and Fraser 2006; Jing et al. 2020; Tanaka and Aihara 2013). If the transmission rate were to be considered constant and only one recovered stage is considered, then the theoretical analysis would show that this kind of models has only a global stable disease-free and endemic equilibrium points (Hethcote et al. 1981; Stech and Williams 1981; El Khalifi and Britton 2023; Guiaş 2023), which does not reflect the seasonal character of the influenza disease. Some exceptions to this situation are some models based on a delay differential equation, where periodic solutions can be obtained (Hethcote 2000), and network models (Acedo et al. 2011), where seasonality arises in a natural way because of the network structure.

In our model, it is clear from Eq. (3) that the transmission rates $\beta_k(t)$ depend upon $\beta(t)$. Based on epidemiological aspects related to seasonal influenza (Ewing et al. 2017; Huang et al. 2023), we propose in this study that $\beta(t)$ should be a sinusoidal periodic function. Its expression is given by

$$\beta(t) = \frac{\beta_0}{2} \left[1 + \cos \left(\frac{2\pi}{T}t - \phi_0 \right) \right] \in [0, \beta_0], \quad (4)$$

where β_0 is the maximum value of the transmission rate, ϕ_0 is the initial phase, and T is the seasonal period, which, based on epidemiological and biological findings, should be around one year (Huang et al. 2023; Lowen and Steel 2014; Thai et al. 2015).

3 Case study: seasonal influenza in Castellón de la Plana city, Valencian Community, Spain

In this section, we use the proposed mathematical modeling approach to describe a real-world case study to explore the applicability and feasibility of the seasonal SIR

model. The case study corresponds to the seasonal influenza between the years 2011 and 2020 in Castellón de la Plana city, in the Valencian Community, Spain. To test the model's ability to represent this scenario, we will fit the model to the available data through a calibration process. It is important to remark that the main motivation of the presentation of the case study is to show the applicability and feasibility of the seasonal SIR_n model to a real-world situation instead of showing a calibration process to estimate the parameters of the seasonal SIR_n model given a real data of influenza. For instance, in Goeyvaerts et al. (2015) a more complex model was fitted to real data and it was found that the model was unidentifiable. Let us proceed to present the case study.

3.1 Data and known parameters

As available calibration data, we have the weekly influenza urgent reported cases in the Castellón de la Plana General Hospital between 2010 and 2020 years, $I_r = \{I_r^{t_i}\}_{i=0}^m$, where $t_0 = 0$ represents the start date, August 9th, 2010 (week 32), and where $t_m = t_{507} = 507$ (weeks) represents the end date, April 27th, 2020 (week 18). It is important to mention that there are available data series of previous years. However, the 2009-2010 season had the entry of a new influenza strain: the A/H1N1 variant (González-Parra et al. 2011; Shubin et al. 2016). In this season, there was a highly atypical epidemic wave, due to a lower level of immunity in the population. In this work, the seasonal SIR_n model has been oriented to medium- to long-term influenza disease, with multiple strains already established in the population and against which the population has acquired a certain immunity. In this case, the alternating circulation of the strains makes that, on average, the magnitude of influenza waves periodically reaches a certain "seasonal steady state". This can be seen in the height of the peaks of reported cases, which are usually similar in different seasons. For this reason, we have chosen a period such as 2010-2020, in which the magnitude of the flu epidemic has been observed regular.

The seasonal influenza data series are shown in Figure 2.

The entire population of Castellón de la Plana is covered by this hospital, so the cases reported are really those of the city in the selected period (private data from the Clinical Documentation and Admissions Department of the Castellón de la Plana General Hospital). Specifically, we know that

- The average annual birth rate in the 2014-2019 period (previous years were not available) was 8.433 births per 1000 inhabitants, i.e., $\lambda = 8.433/(1000 \cdot 52) = 1.623 \cdot 10^{-4} \text{ weeks}^{-1}$ [91]. The average annual death rate in the same period was 8.435 deaths per 1000 inhabitants, i.e., $\mu = 8.435/(1000 \cdot 52) = 1.622 \cdot 10^{-4} \text{ weeks}^{-1}$. With these values, we can assume that the rates are equal and fixed, taking the average, that $\lambda = \mu = 1.622 \cdot 10^{-4} \text{ weeks}^{-1}$. And consequently, we can assume that the size of the total population $N(t) = N$ is constant.
- The average total population size assigned to the hospital is $N = 282\,967$, according to the information we have received from the Clinical Documentation and Admissions Department of the hospital (private communication from Clinical

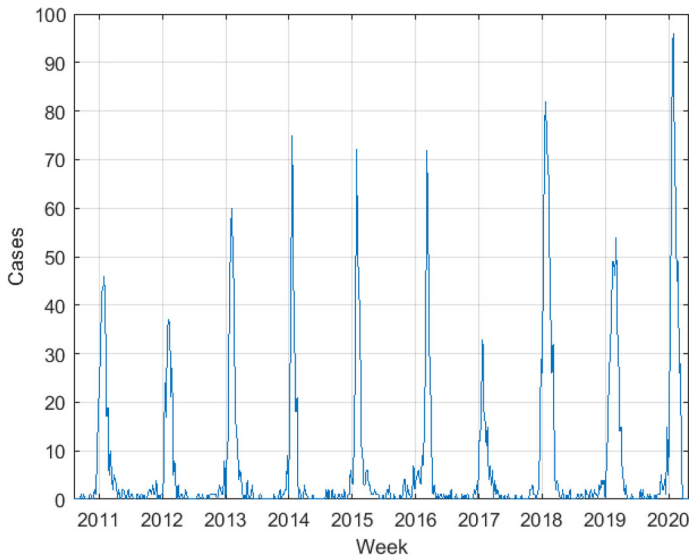


Fig. 2 Seasonal influenza data series: weekly influenza urgent reported cases in the Castellón de la Plana General Hospital in the 2010-2020 period. Source: Clinical Documentation and Admissions Department of the Castellón de la Plana General Hospital (private data)

Documentation and Admissions Department of the Castellón de la Plana General Hospital). Thus, N represents the number of individuals in the wider community who would attend the hospital if they became severely ill.

In addition, we use crucial information (or constraint) for the calibration process: it has been estimated that the number of people who end up getting infected by seasonal influenza each year is between 5 and 15% of the total population [92, (Russell et al. 2008; Stöhr 2002; Tokars et al. 2018)].

Figure 2 shows that every year there is a peak and only one wave per year. Moreover, as expected, this peak occurs over the winter season, although its height varies due to the intrinsic randomness of the virus infection process, its severity on infected individuals and circulating influenza strains (Goeyvaerts et al. 2015). This qualitative behavior is also observed in many other countries around the world (Truscott et al. 2012; Hirve et al. 2016; World Health Organization 2024; Yuan et al. 2021). Using the peak information in the seasonal influenza data series, we have obtained the period T and the initial phase ϕ_0 of the $\beta(t)$ function. First, we have detected the date when the peaks of each season occur, and we have calculated that the average distance between peaks is $T = 52.43$ weeks, i.e., approximately one year, as we already expected. Then, to provide a congruent value of the initial phase ϕ_0 , we have fixed that the transmission rate $\beta(t)$ reaches its maximum (i.e., $\beta(t) = \beta_0$) around the peaks of the seasons, and its minimum (i.e., $\beta(t) = 0$) just at the midpoint between seasons. Therefore, we have taken all the dates located just at the midpoint between two seasons, and we calculated that, on average, the inter-season date (with minimum transmission rate) is around August 6th. Since our series starts on August 9th (practically on the same date, with

only three days of difference), we set $\beta(t_0) = \beta(0) = 0$, and conclude that

$$\beta(0) = \frac{\beta_0}{2} [1 + \cos(-\phi_0)] = 0 \implies \phi_0 = \arccos(-1) = \pi.$$

Therefore, for our case study, we define $\beta(t)$ as

$$\beta(t) = \frac{\beta_0}{2} \left[1 + \cos \left(\frac{2\pi}{52.43} - \pi \right) \right]. \quad (5)$$

For our case study we have proposed a susceptibility function $s(\tau)$ that has an exponential type, with the form

$$s(\tau) = 1 - (1 - s_0)e^{-a\tau} \in [s_0, 1), \quad \forall \tau \geq 0, \quad (6)$$

where $s_0 \in (0, 1]$ represents the initial susceptibility degree after infection and the exponential parameter $a \in \mathbb{R}_+$ modulates the increase in susceptibility, or equivalently, the immunity decay, over time after infection τ . The larger the value of a , the faster the susceptibility increases. Complementary to this function, the immunity degree function can be defined as

$$i(\tau) = 1 - s(\tau) = (1 - s_0)e^{-a\tau} \in [0, 1 - s_0), \quad \forall \tau > 0. \quad (7)$$

Some particular aspects and assumptions related to the chosen susceptibility function $s(\tau)$ are the following:

- Immediately after infection, the individual has an initial degree of susceptibility $s_0 \in (0, 1]$, i.e. $s(0) = s_0$. This degree of susceptibility to a single virus strain of influenza could be set approximately to $s_0 \approx 0$, since immediately after infection, the individual is almost fully protected against this strain. Nevertheless, in the context of seasonal influenza, where there are multiple circulating strains, an individual can become infected by other variants after an infection. It should be taken into account that, after infection, the immune system is very active and there is cross-immunity between strains (El Khalifi and Britton 2023; Goeyvaerts et al. 2015; Xiao and Moghadas 2013). Thus, the susceptibility degree is assumed as $0 < s_0 \leq 1$.
- $s(\tau) \rightarrow 1$ when $\tau \rightarrow \infty$, i.e., as the time after infection increases, the individual tends to become completely susceptible (to lose all immunity). This is related to the fact that the protective immunity wanes and also to the fact that new strains appear which increases the likelihood of becoming infected again (Goeyvaerts et al. 2015; Xiao and Moghadas 2013; Andreasen and Sasaki 2006; De Cezaro et al. 2023; Omori and Sasaki 2013).

For this case study, we choose the value of a such that at 6 years after an individual becomes infected ($\tau = 6$), i.e., 312 weeks after infection (considering 52 weeks per year), the susceptibility degree has recovered to 95% (Xiao and Moghadas 2013;

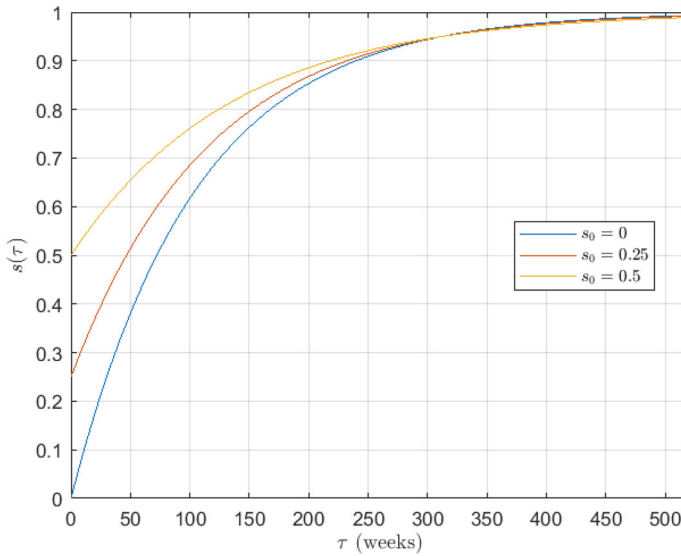


Fig. 3 Susceptibility functions $s_i(\tau)$ for different initial susceptibility degrees s_0 and exponential parameter a

Pitman et al. 2013; Gibson et al. 2016). Thus,

$$s(312) = 1 - (1 - s_0)e^{-a \cdot 312} = 0.95 \implies a = -\frac{1}{312} \log \left(\frac{1 - 0.95}{1 - s_0} \right). \quad (8)$$

We will also consider three scenarios for the initial susceptibility degree s_0 : 0%, 25% and 50%. The susceptibility functions $s_i(\tau)$ for this case study are shown in Figure 3, and are given by

$$s_0 = 0.00, \quad a_1 = 9.602 \cdot 10^{-3} \implies s_1(\tau) = 1 - e^{-9.602 \cdot 10^{-3} \tau}, \quad (9)$$

$$s_0 = 0.25, \quad a_2 = 8.680 \cdot 10^{-3} \implies s_2(\tau) = 1 - 0.75 e^{-8.680 \cdot 10^{-3} \tau}, \quad (10)$$

$$s_0 = 0.50, \quad a_3 = 7.380 \cdot 10^{-3} \implies s_3(\tau) = 1 - 0.50 e^{-7.380 \cdot 10^{-3} \tau}. \quad (11)$$

The partition $\mathcal{T}$ has been applied over a sufficiently large recovery period: 10 years (520 weeks). For this period, we can guarantee that the immunity of a recovered individual would have been reduced to an extremely low level. As the temporal unit of the calibration data is the week, we have defined the partition

$$\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_{520}\} = \{0, 1, \dots, 519\},$$

with $n = 520$ recovered subpopulations, so that

$$\mathcal{S} = \{s(\tau_1), s(\tau_2), \dots, s(\tau_n)\} = \{0, 9.556 \cdot 10^{-3}, \dots, 0.993\}$$

Table 1 Model parameters, with their description, units and values for the case study

Parameter	Description	Units	Value
$\beta(t)$	Transmission rate function	time units ⁻¹	$\frac{\beta_0}{2} \left[1 + \cos \left(\frac{2\pi}{T} t - \phi_0 \right) \right]$
β_0	Amplitude (maximum value) of $\beta(t)$	time units ⁻¹	Unknown
T	Period of $\beta(t)$	time units	52.43 weeks
ϕ_0	Initial phase of $\beta(t)$	adimensional	π
$s(\tau_k)$	Susceptibility degree function	adimensional	$1 - e^{-9.602 \cdot 10^{-3} \tau_k}$
r_k	R_k recovered transition rate	time units ⁻¹	1 weeks ⁻¹
γ	Recovery rate	time units ⁻¹	1 weeks ⁻¹
λ and μ	Birth and death rates	time units ⁻¹	$1.622 \cdot 10^{-4}$ weeks ⁻¹

contains the susceptibility degree of each recovered subpopulation. In particular, note that when an individual arrives at the last recovered subpopulation, it has a susceptibility degree of $s(\tau_{520}) = 0.993 \simeq 1$ (is almost fully susceptible), or an immunity degree of $i(\tau_{520}) \simeq 0$.

With this equidistributed partition $\mathcal{T}$, we can compute the transition rates between the recovered states R_k and R_{k+1} as

$$r_k = \frac{1}{\tau_{k+1} - \tau_k} = 1, \quad \forall k = 1, \dots, n-1.$$

The parameters of model (1), with their description, units and values for the case study, are summarized in Table 1.

3.2 Calibration process

In this subsection, we present the details related to the calibration process of the model to the seasonal influenza data as a real-world case study. The aim of the calibration is to find the estimated numerical values of the unknown model parameters set θ that best fits the available seasonal influenza data. This calibration process is relevant since it allows us to get insight into the seasonal SIR n model dynamics behavior and also to show the applicability of the seasonal SIR n model to the real-world.

In our case study, the unknown model parameter is the amplitude β_0 of the $\beta(t)$ function (see Table 1). The initial conditions of the problem are usually not known. Nevertheless, we are considering for calibration a period in which the magnitude of the epidemic has been similar in all seasons, i.e., it has reached a “seasonal steady state” (as mentioned in Sections 3.1 and 2.2). In order to reach this scenario in the model simulations, first we set the initial conditions to

$$S(0) = N - 1, \quad I(0) = 1, \quad R_1(0) = \dots = R_n(0) = 0.$$

Then, we numerically simulate the model for a long time (warm-up simulation period) until the model reaches a regular periodic behaviour and then obtain the initial conditions when the infected cases are minimal. In our case study, a 20 years warm-up period has been proved enough.

We implement the calibration process using the classical deterministic least square error minimization problem to find the optimal unknown parameters set θ^* that best fits the model to the real-world seasonal influenza data. In our case, the data series is the weekly influenza urgent reported cases (Figure 2). However, it should be noted that this series cannot be directly compared with the infected individuals $I(t) = I(t; \beta_0)$ of the model (1), since this refers to the infected of the entire population, while the data series $\mathbf{I}_r$ only counts the cases reported in the hospital. These cases are those individuals who have been most severely affected by the virus, and who have required hospital medical care. Therefore, we will assume that at week t_i , only a fraction p of the total cases are finally reported. Thus, the reported infected cases of the model can be expressed as

$$I_r(t_i) = p I(t_i), \quad \forall i = 1, \dots, m, \quad (12)$$

where $p \in [0, 1]$ is the reporting fraction, which is also an unknown parameter. At this point, the unknown parameter set θ is defined as

$$\theta = (\beta_0, p) \in \Theta, \quad (13)$$

where $\Theta = \mathbb{R}_+ \times [0, 1]$ is the parameters' search space. Then, the model calibration consists of solving the following least squares minimization problem:

$$\underset{\theta \in \Theta}{\text{minimize}} \sum_{i=1}^m (I_r(t_i) - I_r^i)^2. \quad (14)$$

It is important to point out that the data series represents the new weekly reported cases and $I(t)$ is the number of influenza cases at time t . Since we assume that the average infectious period is 7 days, then $I(t)$ is a good approximation of the new number of cases per week (Chowell 2017). Notice that $I(t)$ is the concurrent infected individuals, which in average stay infectious one week.

As $I(t_i)$ is governed by a nonlinear differential equation, there is no analytical solution for the model (1), so the problem must be solved numerically. Based on the seasonal SIR $_n$ model and the minimization problem, we can apply two key points to the calibration process:

- Although the value provided for the initial phase $\phi_0 = \pi$ is a good approximation, it may not be optimal depending on the value of β_0 . This is because β_0 has the effect of changing the infection force and, consequently, slightly advancing or delaying epidemic waves. Therefore, in order to give more flexibility to the model, we compute after each simulation the discrete cross-correlation function $(\mathbf{I} \star \mathbf{I}_r)(\lambda)$ between the series of total infected $\mathbf{I} = \{I(t_i)\}_{i=1}^m$ and reported $\mathbf{I}_r$ (Rabiner and Gold 1975; Teixeira et al. 2017). Since we know that $\phi_0 \simeq \pi$ is a phase very close to the real one, we have limited the lag λ to 52 weeks (1 year), so that

$\lambda = -52, \dots, 0, \dots, 52$. By doing so, we find the lag λ_0 that causes the maximum cross-correlation, i.e.,

$$\lambda_0 : (\mathbf{I} \star \mathbf{I}_r)(\lambda_0) = \max_{\lambda} \{(\mathbf{I} \star \mathbf{I}_r)(\lambda)\} \quad (15)$$

Finally, prior to the calculation of the least squares function, we redefine

$$I(t) \leftarrow I(t + \lambda_0)$$

so that the offset between the two series is corrected. Indirectly, we are giving the initial phase ϕ_0 some degree of freedom during the calibration process.

- The parameter p does not depend on the model, but acts *a posteriori* to the simulation, scaling the infected data series. Thus, the general solution for this optimization problem in terms of p is given by

$$p = \frac{\sum_{i=1}^m I(t_i) I_r^{t_i}}{\sum_{i=1}^m I(t_i)^2}. \quad (16)$$

Notice that the numerical value of p can be computed with this formula instead of calibrating it. Thus, we reduce the computation time (Andreu-Vilarroig et al. 2024). The parameter p does not affect the dynamics of the model but it depends on the $I(t_i)$ state variable, which is an outcome of the numerical simulation of model (1).

Based on the previous assumptions, the only parameter that is initially estimated is the transmission rate β_0 . The calibration process consists of the following steps:

1. Initially, a sufficiently wide exploration grid for β_0 was set. In our case, we chose the range $[2, 3]$ with 300 points, i.e., a time step of order 10^{-3} .
2. The model is simulated, computing the reporting fraction p and correcting the $I(t)$ function with the lag λ_0 after the simulation.
3. All simulations with an average percentage of new infections per season over the total population P_{new} out of the 5-15% range are discarded. This percentage is computed by numerically solving the differential equation of the cumulative new infected.

$$\dot{I}_{new}(t) = \beta(t) \frac{S(t)I(t)}{N(t)} + \sum_{k=1}^n \beta_k(t) \frac{R_k(t)I(t)}{N(t)} \quad (17)$$

and then computing the average of the percentage of new infections per season as

$$\begin{aligned} P_{new} &= 100 \cdot \frac{1}{m_s} \sum_{i=1}^{m_s+1} \frac{I_{new}(t_{i+1}^s) - I_{new}(t_i^s)}{N} \\ &= 100 \cdot \frac{1}{m_s} \frac{I_{new}(t_{m_s+1}^s) - I_{new}(t_1^s)}{N}, \end{aligned} \quad (18)$$

where $m_s = 10$ is the number of seasons and $\{t_1^s, t_2^s, \dots, t_{m_s+1}^s\}$ are the inter-seasonal times, which mark the beginning (and the end) of each epidemic wave.

4. The sum of squared errors (SSE) in problem (14) is computed for the remaining simulations and the value β_0^* that achieves the minimum SSE is returned as the optimal.

The programming of the model and the calibration method has been carried out in MATLAB. In particular, we have used the *ode45* solver for the numerical resolution of the model.

In the next section we will apply this calibration process for three different assumptions regarding the susceptibility functions defined in Eqs. (9), (10) and (11).

3.3 Results

In this subsection, the results of the SIR_n model fit to the seasonal flu data are shown for the case study after applying the calibration process. The calibration results are shown in Figure 4. We find the best model solution fitted to the data for the three assumed susceptibility functions $s_1(\tau)$, $s_2(\tau)$ and $s_3(\tau)$. Notice that the fitting provides very similar profiles (overlap of curves) despite the use of different susceptibility functions. Table 2, shows the β_0 value range that satisfies the 5-15% seasonal infected restriction, the mean and standard deviation (std.) of the 10% best solutions, the optimal parameter set θ^* and the SSE value for each proposed susceptibility function. Figure 5 shows the SSE versus the different values of β_0 evaluated in the grid search. The SSE is computed by using a total of 520 data points which generates large values for the SSE.

As it can be observed in Figure 4, the calibration process is able to find values of β_0 and p parameters such that the seasonal SIR_n model can approximately describe the real-world seasonal evolution of the influenza disease under different scenarios of susceptibility. Note that despite the forced-seasonality the peaks have different heights due to the gradual waning immunity (Goeyvaerts et al. 2015; Xiao and Moghadas 2013). Fitting a plausible mathematical model to influenza data is very challenging due to the different heights of the peaks (Goeyvaerts et al. 2015). Moreover, oftentimes the models are not identifiable. Several fundamental aspects should be remarked about these results:

1. Note that, in our model, $\mathcal{R}_0$ shows a time-dependent behaviour, since the basal infection rate $\beta(t)$ is sinusoidal. In particular, the basic reproduction number $\mathcal{R}_0(t)$ for the seasonal SIR_n model is given by

$$\mathcal{R}_0(t) = \frac{\bar{\beta}(t)}{\gamma + \mu}, \quad (19)$$

where

$$\bar{\beta}(t) = \frac{1}{n+1} \left(\beta(t) + \sum_{k=1}^n \beta_k(t) \right),$$

Table 2 Calibration results of the seasonal SIR_H model for different susceptibility functions $s(\tau)$: range of β_0 which satisfies the 5-15% seasonal infected restriction, mean $\pm$ std. of the 10% of best solutions θ , best solution θ^* and best SSE

Susceptibility	$s_1(\tau)$ ($s_0 = 0.00$)	$s_2(\tau)$ ($s_0 = 0.25$)	$s_3(\tau)$ ($s_0 = 0.50$)
5-15% restriction β_0 range	[2.200, 2.816]	[2.164, 2.615]	[2.130, 2.328]
10% best sols. $\theta = (\beta_0, p)$ mean $\pm$ std.	$\begin{pmatrix} 2.249 \\ 0.0191 \end{pmatrix} \pm \begin{pmatrix} 0.0642 \\ 3.800 \cdot 10^{-3} \end{pmatrix}$	$\begin{pmatrix} 2.177 \\ 0.0210 \end{pmatrix} \pm \begin{pmatrix} 9.159 \cdot 10^{-3} \\ 1.165 \cdot 10^{-3} \end{pmatrix}$	$\begin{pmatrix} 2.139 \\ 0.0208 \end{pmatrix} \pm \begin{pmatrix} 7.225 \cdot 10^{-3} \\ 1.243 \cdot 10^{-3} \end{pmatrix}$
Best sol. $\theta^* = (\beta_0^*, p^*)$	(2.375, 0.0122)	(2.164, 0.0228)	(2.130, 0.0224)
Best SSE	$4.740 \cdot 10^4$	$4.922 \cdot 10^4$	$4.919 \cdot 10^4$

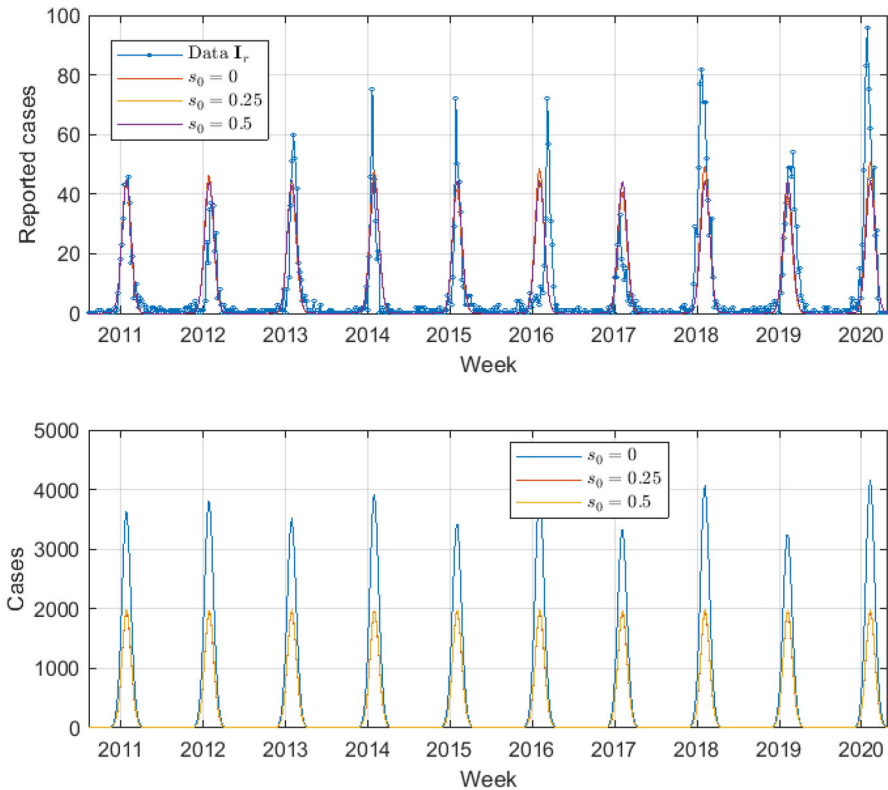


Fig. 4 Best SIR_n model solution. Best fit between the SIR_n model reported infected and data series (top) and model simulation of total infected $I(t)$ (bottom), for different susceptibility scenarios

is the average of all infection rates of susceptible S and recovered classes R_k over a long period of time (Ma and Ma 2005; Goeyvaerts et al. 2015). In this case study, we consider a time unit such as $\gamma + \mu \simeq 1$, so that $\mathcal{R}_0 \simeq \bar{\beta}$. Note that β_0 is the maximum value that $\beta(t)$ can take during the season, so that $\mathcal{R}_0 \simeq \bar{\beta} < \beta_0$ (i.e., β_0 gives an upper bound of the $\mathcal{R}_0$ value). In the calibration results β_0 takes values around 2.25, 2.18 and 2.14 for the three analyzed scenarios. This agrees with what it is usually observed in other influenza mathematical modeling studies, i.e., an $\mathcal{R}_0 < 3$ (Chowell and Brauer 2009; Chowell et al. 2010; Jing et al. 2020; Nikbakht et al. 2019; Samsuzzoha et al. 2013).

2. The reporting fraction p is around 0.02 for the three scenarios, i.e., 2% of the infected cases are urgent reported cases. Equivalently, we can say that, for every urgent reported case, there are $1/p = 50$ unreported infected cases. Note that the unreported cases include the asymptomatic cases that are able to transmit the influenza virus. Note that p does not represent the infection fatality ratio (IFR) nor the case fatality ratio (CFR). However, p must be greater than the IFR since not all the urgent reported cases end up dying. The CFR has been reported as less than 0.1% and it is well-known that it is greater than the IFR (Nishiura 2017;

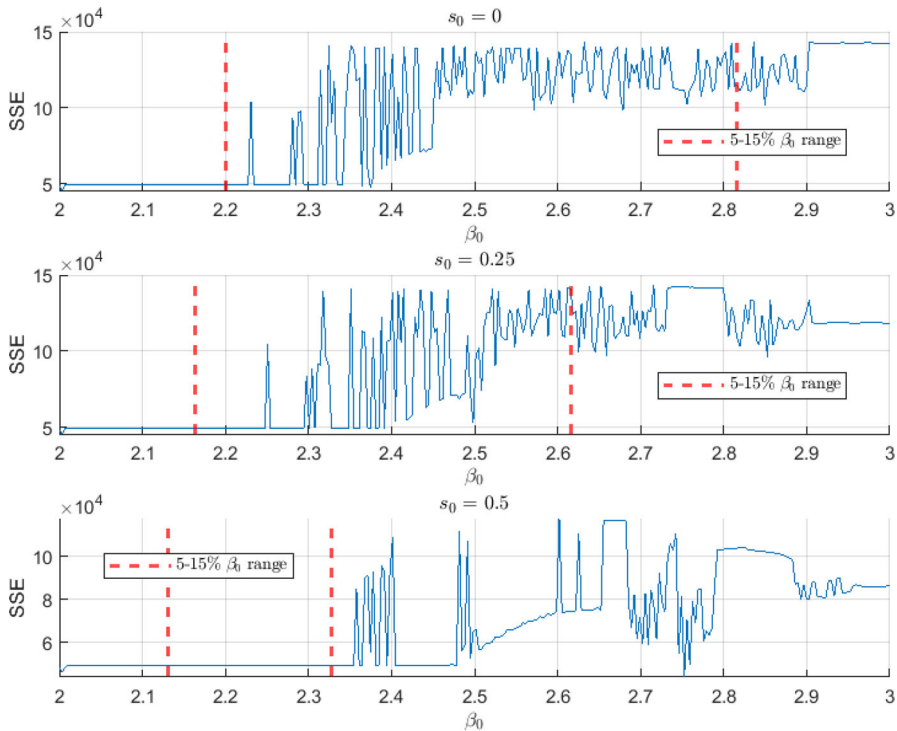


Fig. 5 SSE error function for different β_0 parameter values and different susceptibility scenarios. The red dashed lines delimit the β_0 range that satisfies the 5-15% infected per season restriction

Taubenberger and Morens 2006). Thus, the values obtained for p are in a realistic range.

3. The height of the peaks of years 2011-2013, 2017 and 2019 are well captured. Moreover, the time instants at which practically all the peaks occur are very close to the real ones (about ± 1 -2 week of deviation with respect to the data peaks), despite the multiple transitions of individuals from the recovered n compartments to the infected one. This is a crucial result of the calibration of the model. Having a perfect fit of the height of the peaks is very challenging due to the introduction of new variants, age-structure, and the stochastic factors related to the infectious dynamics (Goeyvaerts et al. 2015; Allen 2008; Arenas et al. 2009).
4. The increase in the initial susceptibility level s_0 implies the reduction of the immunity level, making the susceptibility function closer to $s(\tau) = 1$ (i.e., closer to the SIS model). It can be observed in Table 2 that this increase has the effect of reducing the infection rate β_0 . In other words, to omit the immunity waning causes the immunity to be “transferred” to the infection rate parameter.

Finally, Figure 5 shows the error evolution as a function of β_0 . It can be observed that the function that represents the SSE has large fluctuations and flat regions. The flat regions suggest that the optimization problem is unidentifiable (Kreutz et al. 2012; Raue et al. 2009, 2013). Since in this case we only vary the value of the parameter

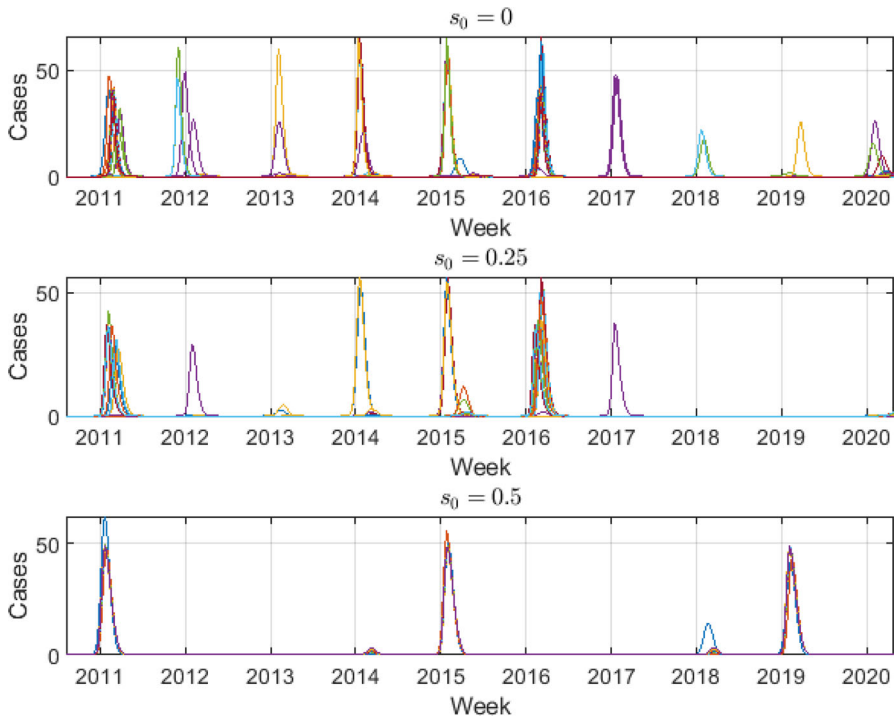


Fig. 6 Unrealistic solutions for different susceptibility scenarios. The solutions shown are the 5% of the numerical simulations with the highest SSE

β_0 we are performing a practical identifiability by using the profile likelihood (Kreutz et al. 2012; Raue et al. 2009, 2013). Note that indirectly, λ_0 gives freedom to the initial phase ϕ_0 during the calibration process. Thus, since β_0 affects the timing of the peak, it is correlated to the phase and, therefore, to λ_0 . Thus, the non-identifiability of the parameters β_0 , p and λ_0 seems reasonable because only two useful pieces of information are available for the calibration process: the series of reported infected and the 5-15% constraint of new seasonal infected (Andreu-Villarraig et al. 2024). However, we have a set of three free parameters in the calibration: β_0 , p and λ_0 . Moreover, the seasonal constraint provides infinitely many possibilities. For the model to be identifiable, additional information should be incorporated in the calibration process (Raue et al. 2009, 2013; González-Parra et al. 2018). Nevertheless, by looking at the solutions where the SSE is large we have found that those model solutions would yield unrealistic solutions, where several season peaks disappear in the infected curves (see Figure 6). Thus, although in this problem we do not have identifiable parameters, we have found ranges for β_0 that satisfy the 5-15% constraint and that provide realistic feasible solutions. Thus, it is shown that obtaining a set of parameters' values of the seasonal SIRn model such that feasible solutions are obtained is challenging (Goeyvaerts et al. 2015).

4 Discussion

The mathematical modeling of the dynamics of influenza at the population level is extremely challenging due to two main factors. The first one is the seasonality nature of the influenza virus. The second one is waning immunity. Mainly, these two factors have been considered separately in epidemiological mathematical modeling. In this study, we have combined these main factors into a mathematical model. Oftentimes seasonality has been modeled by defining the influenza transmission rate as a periodic function, with higher values in winter seasons (Brauer et al. 2019; Goeyvaerts et al. 2015). The mathematical model presented in this work is in some way versatile since the time of the peak and the pattern of the epidemic wave can be modified by changing the values of some parameters of the model. This is important since different regions or countries might have slight differences in the influenza seasonal waves (Goeyvaerts et al. 2015). Different circulating influenza strains and climatic factors can create variations in the pattern of the waves.

Regarding the mathematical analysis of the proposed mathematical model it can be proven that under some conditions there are periodic solutions, but a full mathematical analysis is out of the scope of this article. The only expected steady states are the periodic solutions and a disease-free equilibrium point. Another mathematical aspect of the model is that as $n \rightarrow \infty$, the number of recovered stages in the SIR_n model increases and then there is a continuity of recovered stages. Thus, this asymptotic model is given by an ODE-PDE system where further mathematical analysis can be done (El Khalifi and Britton 2023; Inaba 2001).

As in any modeling process, our proposed mathematical model is not a fully exhaustive modeling approach that includes all the details of the real-world. However, it results in a much more detailed model than a traditional epidemiological compartmental model such as SIR or SEIR models. The subdivision of the recovered state into n multiple stages, while increasing complexity, introduces richer dynamics and allows for a more realistic description of seasonal influenza by incorporating the immunity effect (El Khalifi and Britton 2023; Hethcote et al. 1981). It is important to emphasize that our proposed model considers implicitly natural cross-immunity since the decay of immunity can occur over several seasons so that the recovered population is partially protected against reinfection (Truscott et al. 2012; Xiao and Moghadas 2013). It should be noted that, while a simpler model, such as a classic SIR framework without waning immunity, could potentially achieve a similar fit to the observed data, our primary objective extended beyond purely predictive accuracy. By explicitly modeling waning immunity through the susceptibility function $s(\tau)$, we were able to disentangle its effects from the basal infection rate β_0 , providing a more precise estimate of the virus transmission strength. For instance, as mentioned in Section 3.3 (point 4), omitting immunity would effectively “transfer” its impact to the β_0 parameter (by reducing it), potentially underestimating the virus infectiousness. This fact also affects to other parameters of great medical interest, such as the basic reproduction number $\mathcal{R}_0$. Thus, while a simpler models (e.g., SIR, SLIR, etc.) might predict trends similarly, our approach distinguishes the virus’s inherent infectiousness, a key insight for understanding its spread. A detailed statistical comparison with simpler models, though possible, falls outside this study’s scope, as our goal was not to optimize pre-

dictive accuracy but to provide epidemiologically meaningful parameters, especially since seasonal flu trends are already predictable with basic models once cases emerge.

To the best of our knowledge, the idea of taking into account both forced-seasonality and explicit gradual waning immunity has only been considered in a few papers, despite both factors play a crucial role in the seasonal influenza dynamics (Dafilis et al. 2014; Xiao and Moghadas 2013). Moreover, our proposed model is different than the ones presented in those works. The mathematical modeling approach proposed in this study might not be appropriate for describing the dynamics of influenza in tropical regions. In Yuan et al. (2021) it was found that a model that incorporates the effect of both humidity and temperature was able to describe the influenza epidemic patterns in Hong Kong. Their results suggested that a shorter immunity period can be used to simulate the co-circulation of influenza virus strains. Another interesting work proposed an agent-based model to describe the A/H1N1 influenza over one season. The model combines seasonality and viral mutation in an influenza pandemic and assumes that the $\mathcal{R}_0$ value varies over time by using a sinusoidal function (Shi et al. 2010).

As in any mathematical model regarding epidemics or infectious diseases, there are limitations. For instance, the constructed model does not consider the vaccination of the population explicitly. However, the model considers that since the influenza vaccination plans are implemented every year and influenza viruses have been circulating for many years then the influenza dynamics is in a steady state scenario with a seasonal behavior due to the main factors explained in this study. Thus, the vaccinated people are distributed in the different recovered subpopulations with variable immunity. The proposed mathematical modeling approach does not differentiate people with natural or induced immunity. Thus, their susceptibilities vary at the same rate over time (see Goeyvaerts et al. (2015)). We assume that the effect of this approximation is relatively small in comparison with other main factors that we have considered, such as seasonality and gradual waning immunity. The cross-immunity is considered implicitly in the susceptibility function and can be adjusted depending on the circulating influenza strains. However, the seasonal SIR n model does not consider explicitly subpopulations with different levels of protective cross-immunity.

Another important aspect that requires careful attention is the time after infection τ . It seems plausible that an older person that have contracted the influenza virus several times might have a different immunological profile to individuals in other age groups. In the seasonal SIR n model there is no age-structure. Nevertheless, a potential model designed with different infectious stages depending on age might be reasonable, even though this would increase the complexity of the model (Goeyvaerts et al. 2015). There are other limitations that are more general to mathematical models based on ODEs such as the homogeneous mixing assumption (Brauer et al. 2019; Hethcote 2000). For more details about limitations of this type of mathematical models see (Goeyvaerts et al. 2015).

Recently, in El Khalifi and Britton (2023) the authors developed two mathematical models where individual immunity wanes gradually by means of linear or exponential functions. Thus, the classical SIR model is extended by including several recovered compartments. In other two works the authors have considered different levels of susceptibility of the individuals (Guias 2023, ?). Nevertheless, the seasonality effect observed in influenza has not been included (Foxman et al. 2015). On the other hand,

there are other works that have included seasonally-forced functions in order to deal with the seasonality of influenza, but which do not include explicit gradual waning immunity (Arenas et al. 2021; Edlund et al. 2011; Evans et al. 2005; He et al. 2013; Malek and Hoque 2024; Mahmud et al. 2023; Tanaka and Aihara 2013). There are other approaches to explain the seasonality and epidemic waves with mathematical models (González-Parra et al. 2011; Stech and Williams 1981; Yang and Jin 2021). Another aspects that could be taken into account are climate factors (Gashaw et al. 2019; González-Parra et al. 2016; Hirve et al. 2016; Moore et al. 2014; Thai et al. 2015; Yuan et al. 2021) and how they can impact the influenza seasonality. In addition, studies that provide genomic sequence analysis of the influenza viruses are useful to understand the dynamics of influenza for different discrete outbreaks (Rambaut et al. 2008).

5 Conclusions

In this paper, we presented a mathematical model, the seasonal SIR_n model, that integrates forced-seasonality and gradual waning immunity. The progressive waning immunity has been introduced into the seasonal SIR_n model structure by considering multiple recovery stages with attenuated transmission rates by a susceptibility factor. To show the applicability and reliability of the seasonal SIR_n model to a real-world case, we have carried out a model calibration with a data series of influenza infections reported in the 2010–2020 period at the General Hospital of Castellón de la Plana, Spain. The calibration results show that the seasonal SIR_n model is able to approximately simulate and replicate the reported infected data. The obtained results allowed us to estimate the ranges for the numerical values of the basic reproduction number $\mathcal{R}_0$, which are in agreement with previous results from the scientific literature related to influenza. It is important to remark that the results were obtained under different immunity decay scenarios. The model shows (through the immunity process) that the epidemic in one year is dependent on the previous years, e.g., a low epidemic in year n results in a higher epidemic in year $n + 1$ because of lower immunity generated in year n . However, since the influenza virus and its variants have been present in the human population for a long time, we assumed a steady state scenario and fitted it using the seasonal SIR_n model. We used scientific data to select different waning immunity forms (El Khalifi and Britton 2023; Ranjeva et al. 2019; Xiao and Moghadas 2013). The specific forms that we implemented for the waning immunity have not been used previously.

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Declarations

Declaration of interest None

Ethics The data relating to the patients have been properly anonymized in accordance with General Data Protection Regulation (EU) 2016/679. For this study informed consent has been waived by the Hospital General Universitario de Castellón ethics committee due to the anonymity and retrospective nature of the study.

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